Unive Mentorship Program Summer 2026
Project Proposal

Configurable ECG Peak Detection and Heart Rate Estimation IP Core

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a. Project Title

Configurable ECG Peak Detection and Heart Rate Estimation IP Core

b. Project Objective

The objective of this project is to design and implement a digital hardware IP core that detects R-peaks in incoming ECG (Electrocardiogram) signals and estimates the patient's heart rate in real time. The design will be developed in Verilog HDL and implemented using the OpenROAD RTL-to-GDSII flow on the IHP SG13G2 (130 nm) technology. The project aims to demonstrate how dedicated hardware accelerators can provide fast, low-power, and reliable biomedical signal processing for wearable and portable healthcare devices.

c. Problem Statement

Cardiovascular diseases remain one of the leading causes of death worldwide. Continuous ECG monitoring enables early detection of cardiac abnormalities, but software-based processing on general-purpose processors may introduce latency and increase power consumption. A dedicated hardware IP core for ECG peak detection can process ECG signals efficiently with low latency and reduced power consumption, making it suitable for wearable health monitors, bedside patient monitoring systems, and portable medical devices.

d. Application Domain

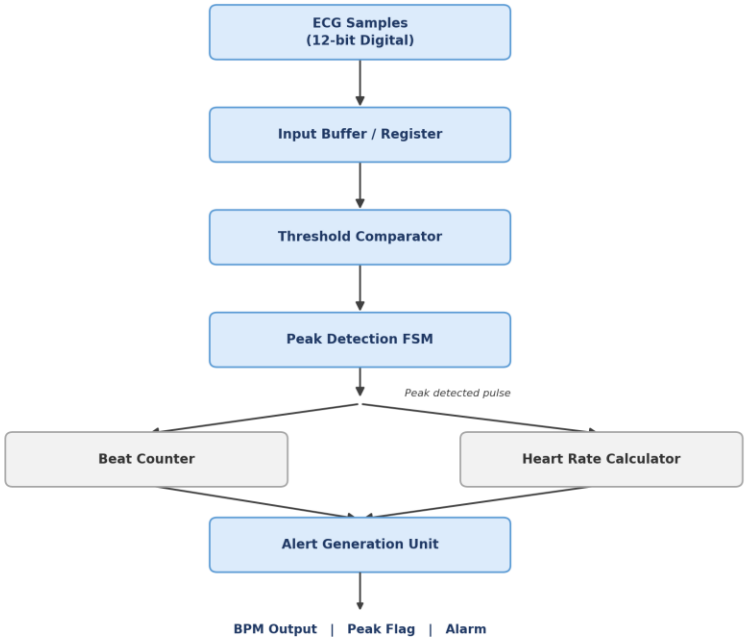
Healthcare & Biomedical

e. Project Overview

This project implements a configurable digital IP block that receives digitized ECG samples and detects the R-peaks of the ECG waveform using threshold-based peak detection logic. The detected peaks are counted over a predefined time window to estimate the heart rate in Beats Per Minute (BPM). The module also generates alert signals if the heart rate falls outside programmable normal limits.

The design is fully synchronous and modular, allowing each block to be independently verified and synthesized. It is intended for ASIC implementation using the OpenROAD flow.

f. Block Diagram



g. Features

- Processes 12-bit digital ECG input samples
- Real-time R-peak detection
- Heart rate estimation (BPM)
- Configurable detection threshold
- Programmable tachycardia and bradycardia limits
- Fully synchronous digital design
- Finite State Machine (FSM)-based control
- Parameterizable architecture
- Low-area implementation
- Compatible with OpenROAD ASIC flow
- Modular RTL architecture
- Easy integration with biomedical monitoring systems

h. Design Specifications

Parameter	Specification
Technology	IHP SG13G2 (130 nm)
Hardware Description Language	Verilog HDL
Design Flow	OpenROAD RTL-to-GDSII
Clock Frequency	50 MHz

Input Data Width	12 bits
Reset	Active-high synchronous reset
Clock	Single clock domain
Estimated Inputs	5
Estimated Outputs	4

Inputs

Signal	Width	Description
clk	1	System clock
rst	1	System reset
ecg_sample	12	Digital ECG sample
sample_valid	1	Sample valid signal
threshold	12	Configurable detection threshold

Outputs

Signal	Width	Description
peak_detected	1	R-peak detection pulse
bpm	8	Estimated heart rate
alert	1	Heart rate abnormal alert
data_valid	1	Output valid signal

Expected Deliverables

- Verilog RTL
- Testbench
- Simulation waveforms
- SDC constraints
- config.mk
- Synthesis reports
- Floorplan
- Placement
- Clock Tree Synthesis (CTS)
- Routing
- Timing report
- Area report
- Power report

- Final GDSII layout
- Technical documentation